Project Planning and Project Estimation Techniques

Employees in S/W industry at Managerial level are the most experienced persons. They often use different estimations techniques for proper planning to make optimum use of available resources during software development.

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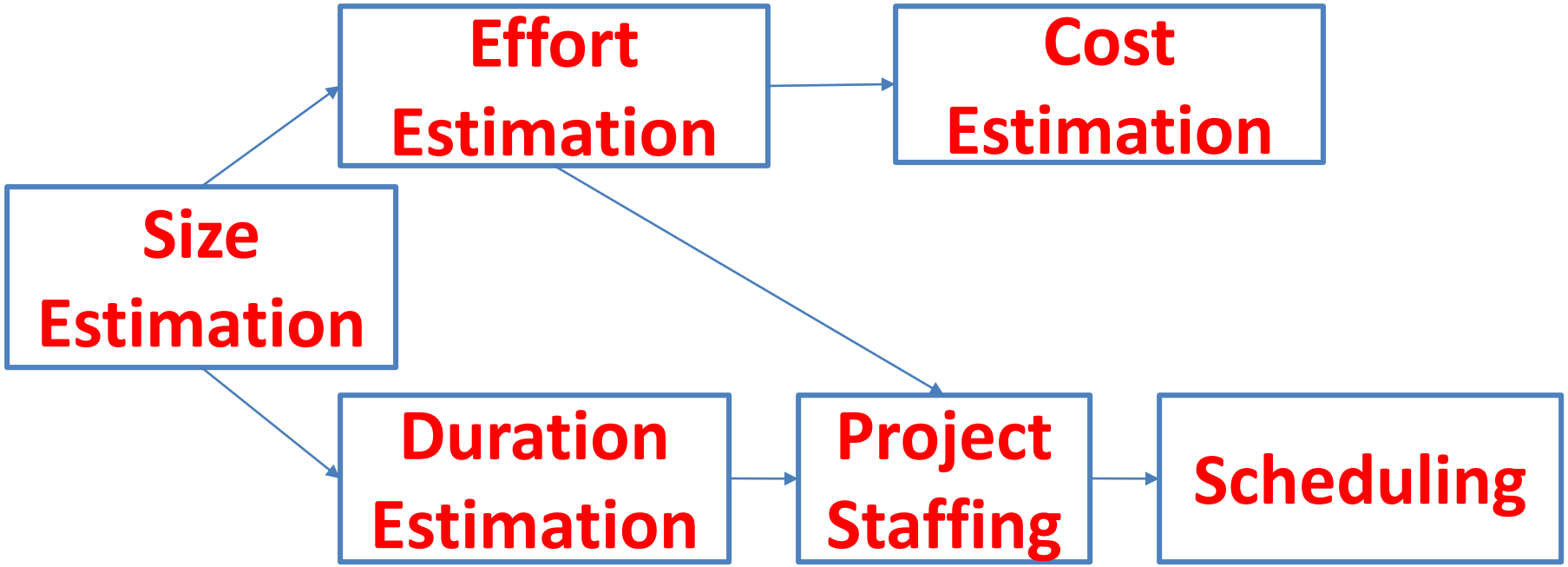
Responsibilities of a software project manager

- Software **project managers** take the overall responsibility for success of a project. **Manager** is responsible for project **planning**, and project **monitoring** and **control** activities.
- Some of the visible **job responsibility** of a **project manager**:
  - project proposal writing,
  - project cost estimation,
  - scheduling,
  - project staffing,
  - software process tailoring,
  - project monitoring and control,
  - software configuration management,
  - risk management,
  - interfacing with clients,
  - managerial report writing and presentations, etc.

Project planning

- **Once a project is found to be feasible**, software **project managers undertake project planning before starting of any development activity**. Project planning consists of:
  - Estimating the following attributes of the project
    - **Project size**: What will be problem complexity in terms of the effort and time required to develop the product?
    - **Cost**: How much is it going to cost to develop the project?
    - **Duration**: How long is it going to take to complete development?
    - **Effort**: How much effort would be required?
  - The subsequent planning activities such as **scheduling of manpower and other resources**, **staff organization**, **risk identifications**, miscellaneous plans are based on the accuracy of these estimations.

Precedence ordering among planning activities



Before proceeding for different estimations, **FIRST we need to have an idea of the product size** to develop. **Size estimation** (**Without looking at the product**) **is very difficult**. **But managers with sufficient experience in similar s/w development can make a fair guess**. Once size is estimated, other activities can be systematically completed.

Sliding Window Planning

- **Project planning** is a **very challenging** activity. **Improper planning** **can** even **cause project failure**.
- Especially for large projects, it is very much difficult to make accurate plans.
- A part of this difficulty is due to the fact that the proper parameters, scope of the project, project staff, etc. may change during the span of the project.
  - In order to overcome this problem, sometimes project managers undertake **project planning in stages**. Planning a project over a number of stages protects managers from making big commitments too early. This technique of staggered planning is known as **Sliding Window Planning**.
  - In this technique, starting with an initial plan, the project is planned more accurately in successive development stages.

Metrics for software project size estimation

- Accurate **estimation of the problem size is fundamental to satisfactory estimation of effort, time duration and cost of a software project**.
- In order to be able to accurately estimate the project size, some important metrics should be defined in terms of which the project size can be expressed.
  - The **size of a problem** is **neither** the **source code size** in terms of **number of bytes**, **nor** the **byte size** of the **executable code**.
  - The **project size is a measure of the problem complexity in terms of the effort and time required to develop the product**.
  - Two popular metrics used to estimate size are: **lines of code (LOC)** and **function point (FP)**

Lines of Code (LOC)

- LOC is the simplest and most popular metrics available to estimate project size.
- The **project size is estimated by counting the number of source instructions in the developed program**.
  - **Comment statements and header lines ignored in count**.
- Determining the **LOC count at the end** of a project **is simple**. However, **accurate estimation** of the **LOC count at the beginning** of a project **is very difficult**.
- In order to **estimate the LOC** count **at the beginning** of a project we need to:
  - **divide the problem into modules**, and each **module into sub-modules** and so on, until the sizes of the different **leaf-level modules can be approximately predicted**.
  - may be past experience in developing similar products will be helpful.
  - Estimate **total project size by combining estimation of the lowest level modules**

Shortcomings of LOC metric

- **LOC** gives a numerical value of **problem size** that can **vary widely with individual coding style**
  - For example, one programmer might write several source instructions on a single line whereas another might split a single instruction across several lines.
  - This problem may be overcome by counting the language tokens in the program rather than the lines of code.
- A good problem size **measure should consider the overall complexity of the problem and the effort needed** to solve it. That is, **it should consider** the local **effort needed to specify, design, code, test**, etc. and **not just the coding effort**.
- **LOC** measure **correlates poorly with the quality and efficiency of the code**. **Larger code size does not necessarily imply better quality or higher efficiency**.

Shortcomings of LOC

- **LOC** metric **penalizes use of higher-level programming languages, code reuse**, etc.
  - if a programmer uses several library routines, then the LOC count will be lower.
- **LOC** metric **measures the lexical complexity of a program and does not address the more important but subtle issues of logical or structural complexities**.
  - Between two programs with equal LOC count, a program having complex logic would require much more effort to develop than a program with very simple logic.

Shortcomings of LOC

- It is very **difficult to accurately estimate LOC in the final product from the problem specification**.
  - The LOC count can be accurately computed only after the code has been fully developed.
- Therefore, the **LOC metric is of little use to the project managers during project planning**, since project planning is carried out even before any development activity has started.
  - This possibly is the **biggest shortcoming of the LOC metric from the project manager's perspective**.

Function point (FP) proposed by Albrecht in 1983

- This metric overcomes many of the shortcomings of the LOC metric.
- One **important advantages** of using **FP** metric is that it **can be used to easily estimate the size of a software product directly from the problem specification**, whereas the size can be accurately determined only after the product has fully been developed in LOC count.
- The **conceptual idea behind the FP metric** is that **the size of a software product is directly dependent on the number of different functions or features it supports and technical issues in implementation**.

Function point (FP) proposed by Albrecht in 1983

- Broadly **software development** can be viewed as development of **five features** such as **(i)** **input** components, **(ii)** **output** components, **(iii)** **query** components, **(iv)** **Files/Database** components and **(v)** **User interfaces**.
- **Size of the software can be estimated** depending upon amount of these features to be developed.
- Depending on the **difficulty** in developing these features **weights** are associated with these features.

Function point (FP) proposed by Albrecht in 1983

- The size **of a product** in function points (**FP**) can be expressed as the **weighted sum** of **five problem characteristics: (Number of inputs), (Number of outputs), (Number of inquiries), (Number of files), (Number of interfaces)**
- **Function point computation is a two step process**.
- The **first step** is to compute the **unadjusted function point (UFP)** using formula:

  UFP = (Number of inputs)*4 + (Number of outputs)*5 + (Number of inquiries)*4 + (Number of files)*10 + (Number of interfaces)*10

  – The **weights associated** with the five characteristics were **proposed empirically** and **validated by the observations over many projects**.

Refinement in weights assigned to Function point (FP) entities

Type	Simple	Average	Complex
inputs	3	4	6
outputs	4	5	7
inquiries	3	4	6
files	7	10	15
interfaces	5	7	10

These weights can vary among software. If we can categories software in to three types, simple, average and complex, then weight associated with different software components are in order simple<average<complex.

Function point (FP) proposed by Albrecht in 1983

- Once the unadjusted function point (UFP) is computed, the **technical complexity factor (TCF)** is computed next **in 2nd step**.
- **TCF** refines the UFP measure by considering **fourteen other factors** such as high **transaction rates**, **throughput**, and **response time** requirements, **etc**. **Each of these 14 factors is assigned from 0 (not present or no influence) to 5 (strong influence)**. The resulting numbers are summed, yielding the total **degree of influence (DI)**.
- Now, **TCF is computed as (0.65+0.01*DI)**. As **DI can vary from 0 to 70, TCF can vary from 0.65 to 1.35**.
- **Finally, FP=UFP*TCF**.

Function point (FP) proposed by Albrecht in 1983

- **Number of inputs:** Each data item input by the user is counted. Individual data items input by the user are not considered in the calculation of the number of inputs, but **a group of related inputs are considered as a single input**.
  - For example, while entering the data concerning an employee to an employee pay roll software; the data items name, age, sex, address, phone number, etc. are together considered as a single input. All these data items can be considered to be related, since they pertain to a single employee.

Function point (FP) proposed by Albrecht in 1983

- **Number of outputs:** The outputs considered refer to **reports** printed, **screen outputs**, **error messages** produced, etc. While outputting the number of outputs the individual data items within a report are not considered, but a set of related data items is counted as one input.
- **Number of inquiries:** Number of inquiries is the number of **distinct interactive queries** which can be made by the users. These inquiries are the user commands which require specific action by the system.
- **Number of files: Each logical file** is counted. A logical file means groups of **logically related data**. Thus, logical files can be **data structures** or **physical files**.
- **Number of interfaces:** Here the interfaces considered are the **interfaces used to exchange information** with other systems. Examples of such interfaces are **data files on tapes**, disks, **communication links** with other systems etc.

Shortcomings of Function point (FP)

- A **major shortcoming** of the function point measure is that it **does not take into account the algorithmic complexity of a software**.
  - That is, the **function point metric assumes** that the **effort required to design and develop any two functionalities of the system is the same**.
- **It only takes the number of functions** that the system supports into consideration **without distinguishing the difficulty level of developing** the various functionalities.
- To overcome this problem, **an extension of the function point metric** called **feature point metric** is proposed.

Feature point

- Feature point metric incorporates an extra parameter **algorithm complexity,** in function point measure
- This parameter ensures that the computed size using the feature point metric reflects the fact that the more is the complexity of a function, the greater is the effort required to develop it and therefore its size should be larger compared to simpler functions.